

# Product Laws of Exponents

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## MA.7.NSO.1.1

*I can use the product of powers, power of a product, power of a power and identity laws to rewrite and evaluate a numerical expression.*

*Last unit you took roots apart.*

**Today you put powers together — and the base has to match.**

# Warm-Up

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*On your own. Four minutes. No notes.*

1.  $\sqrt{81}$

2. Between which two whole numbers is  $\sqrt{40}$  ?

3. Solve  $3x - 7 = 11$

4. Evaluate  $\left(\frac{2}{3}\right)^2$

# Warm-Up

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*Answers.*

1.  $\sqrt{81}$      **9**

2. Between which two whole numbers is  $\sqrt{40}$ ?     **6 and 7**

3. Solve  $3x - 7 = 11$       **$x = 6$**

4. Evaluate  $\left(\frac{2}{3}\right)^2$       **$\frac{4}{9}$**

# Notes

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*Copy the three words.*

## I. A power is a shorthand for repeated multiplication

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A. **base** — the number used as a factor

B. **exponent** — how many times the base is a factor

C. **power** — the whole thing, base and exponent together

$$2^3 = 2 \cdot 2 \cdot 2 = 8 \qquad \left(\frac{2}{3}\right)^2 = \frac{2}{3} \cdot \frac{2}{3} = \frac{4}{9}$$

# Notes

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*Where the rule comes from.*

## II. Product of powers — same base, add the exponents

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$$3^2 \cdot 3^3 = (3 \cdot 3)(3 \cdot 3 \cdot 3) = 3^5$$

$$3^2 \cdot 2^3 = (3 \cdot 3)(2 \cdot 2 \cdot 2) \quad \text{— different bases, no shortcut}$$

**Product of powers:**  $a^m \cdot a^n = a^{m+n}$

Check with values:  $3^2 \cdot 3^3 = 9 \cdot 27 = 243 = 3^5$

# Notes

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## III. Power of a product — the exponent goes to every factor

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$$(2 \cdot 5)^3 = (2 \cdot 5)(2 \cdot 5)(2 \cdot 5) = (2 \cdot 2 \cdot 2)(5 \cdot 5 \cdot 5) = 2^3 \cdot 5^3$$

**Power of a product:**  $(ab)^m = a^m \cdot b^m$

Works for a product only, never for a sum:

$$(10 + (-2))^3 = 8^3 = 512 \quad \text{but} \quad 10^3 + (-2)^3 = 992$$

# Notes

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## IV. Power of a power, and the exponent 1

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| Power of a power | Repeated multiplication                 | Single power |
|------------------|-----------------------------------------|--------------|
| $(4^2)^3$        | $4^2 \cdot 4^2 \cdot 4^2$               | $4^6$        |
| $(10^4)^4$       | $10^4 \cdot 10^4 \cdot 10^4 \cdot 10^4$ | $10^{16}$    |
| $(3^1)^5$        | $3 \cdot 3 \cdot 3 \cdot 3 \cdot 3$     | $3^5$        |

**Power of a power:**  $(a^m)^n = a^{mn}$       **Identity:**  $a^1 = a$

# Example 1

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*Two ways to the same value.*

$$3^2 \cdot 3^3$$

**Which law? Then find the value.**

# Example 1

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*Worked both ways.*

$$3^2 \cdot 3^3 = 3^{2+3} = 3^5 \quad \text{product of powers: same base, add}$$

$$3^5 = 243 \quad \text{evaluate the single power}$$

$$9 \cdot 27 = 243 \quad \text{check: evaluate each, then multiply}$$

**Answer: 243**

# Your Turn

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*Same steps, your numbers. Boards up when done.*

$$2^3 \cdot 2^4$$

# Your Turn

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*Answer.*

$$2^3 \cdot 2^4$$

same base, add:  $2^7$

$$\textbf{Answer: } 2^7 = 128$$

# Example 2

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*A negative base and a power of a power.*

$$[(-2)^3]^2$$

**Which law? Then find the value.**

# Example 2

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*Worked.*

$$[(-2)^3]^2 = (-2)^{3 \cdot 2} = (-2)^6 \quad \text{power of a power: multiply}$$

$$(-2)^6 = 64 \quad \text{six negative factors — positive}$$

$$(-8)^2 = 64 \quad \text{check: } (-2)^3 = -8, \text{ then square}$$

**Answer: 64**

# Your Turn

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*Same steps, your numbers. Boards up when done.*

$$[(-3)^2]^3$$

# Your Turn

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*Answer.*

$$[(-3)^2]^3$$

multiply the exponents:  $(-3)^6$

**Answer:  $(-3)^6 = 729$**

# Whiteboards • Question 1 of 9

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*Boards up on three.*

$$5^2 \cdot 5^3$$

**Answer it.**

*Write it as one power.*

# Whiteboards • Question 1 of 9

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*Answer.*

$$5^2 \cdot 5^3$$

same base, so add the exponents

**Answer:  $5^5$**

# Whiteboards • Question 2 of 9

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*Boards up on three.*

A rectangle is  $2^4$  centimeters long and  $2^2$  centimeters wide.

Its perimeter is 40 centimeters.

**What is the area of the rectangle, in square centimeters?**

**Answer it.**

# Whiteboards • Question 2 of 9

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*Answer.*

A rectangle is  $2^4$  centimeters long and  $2^2$  centimeters wide.

Its perimeter is 40 centimeters.

**What is the area of the rectangle, in square centimeters?**

$$2^4 \cdot 2^2 = 2^6$$

**Answer:  $2^6 = 64$  square centimeters**

# Whiteboards • Question 3 of 9

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*Boards up on three.*

$$(2 \cdot 5)^3$$

**Answer it.**

*Find the value.*

# Whiteboards • Question 3 of 9

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*Answer.*

$$(2 \cdot 5)^3$$

$$2^3 \cdot 5^3 = 8 \cdot 125, \text{ or } 10^3$$

**Answer: 1000**

# Whiteboards • Question 4 of 9

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*Boards up on three.*

$$(3^2)^4$$

**Answer it.**

*Write it as one power.*

# Whiteboards • Question 4 of 9

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*Answer.*

$$(3^2)^4$$

four copies of  $3^2$ , so eight 3s

**Answer:  $3^8$**

# Whiteboards • Question 5 of 9

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*Boards up on three.*

$$4^3 \cdot 4^2$$

Which expression is equivalent?

**A.**  $16^5$

**B.**  $8^5$

**C.**  $4^6$

**D.**  $4^5$

**Answer it.**

# Whiteboards · Question 5 of 9

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*Answer.*

$$4^3 \cdot 4^2$$

Which expression is equivalent?

**A.**  $16^5$

**B.**  $8^5$

**C.**  $4^6$

**D.**  $4^5$

**Answer: D —  $4^5$**

# Whiteboards • Question 6 of 9

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*Boards up on three.*

$$(2^3)^2$$

Which expression is equivalent?

**A.**  $2^5$

**B.**  $4^6$

**C.**  $2^6$

**D.**  $2^9$

**Answer it.**

# Whiteboards • Question 6 of 9

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*Answer.*

$$(2^3)^2$$

Which expression is equivalent?

**A.**  $2^5$

**B.**  $4^6$

**C.**  $2^6$

**D.**  $2^9$

**Answer: C —  $2^6$**

# Whiteboards • Question 7 of 9

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*Boards up on three.*

$$(-3)^2 \cdot (-3)^1$$

**Answer it.**

*Find the value.*

# Whiteboards • Question 7 of 9

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*Answer.*

$$(-3)^2 \cdot (-3)^1$$

$(-3)^3$  — three negative factors

**Answer: -27**

# Whiteboards • Question 8 of 9

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*Boards up on three.*

$$\left(\frac{1}{2}\right)^2 \cdot \left(\frac{1}{2}\right)^3$$

**Answer it.**

*Find the value.*

# Whiteboards • Question 8 of 9

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*Answer.*

$$\left(\frac{1}{2}\right)^2 \cdot \left(\frac{1}{2}\right)^3$$
$$\left(\frac{1}{2}\right)^5$$

**Answer:**  $\frac{1}{32}$

# Whiteboards • Question 9 of 9

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*Take your time. Boards up when you have written it.*

Aiyanna says she can use the power of a product law to evaluate  $(10 + (-2))^3$ . Is she right?

**Write your answer in sentences.**

*Say what the law needs, and show the value.*

# Whiteboards • Question 9 of 9

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*Answer.*

Aiyanna says she can use the power of a product law to evaluate  $(10 + (-2))^3$ . Is she right?

the parentheses hold a SUM, not a product

**Answer: No —  $(10 + (-2))^3 = 8^3 = 512$**

# Independent Set

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*Six questions. On your own, in silence.*

1. Six questions. Work down the page.
2. Show the step that does the work, not just the answer.
3. Silence until the six minutes are up.

# IXL

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*Last five minutes.*

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Every skill listed is required, to a SmartScore of 67. Due at the start of the next class.

## **Today's skills — all required**

Evaluate powers — SZQ

Multiplication with exponents — 5SK

Power rule — 8DA

Understanding exponents — BFA